AI Project Documentation

Fake vs Real Image Detection

Using Transfer Learning (EfficientNetB0 + TensorFlow/Keras)

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1. Introduction

1.1 Background

The rapid advancement of Generative Artificial Intelligence — particularly Generative Adversarial Networks (GANs) and Diffusion Models — has made it increasingly easy to create photorealistic images that are indistinguishable from genuine photographs to the human eye. These AI-generated images pose significant threats in areas such as misinformation, media forgery, identity fraud, and social engineering attacks.

Detecting such fake images has therefore become a critical research problem in computer vision and AI safety. Traditional methods relying on handcrafted features are no longer sufficient. Deep learning-based approaches, especially using Transfer Learning, offer a powerful and efficient solution to this challenge.

1.2 Problem Statement

Given an input image, the goal is to classify it as either:

- REAL — a genuine photograph captured by a camera
- FAKE — an image synthetically generated by an AI model (GAN-generated)

This is formulated as a binary image classification problem, solved using Transfer Learning with a deep convolutional neural network pretrained on ImageNet.

1.3 Objectives

- Apply Transfer Learning using EfficientNetB0 pretrained on ImageNet
- Fine-tune the model on the CIFAKE dataset (60,000 real + 60,000 AI-generated images)
- Achieve high classification accuracy on unseen test images
- Demonstrate the two-phase training strategy: feature extraction followed by fine-tuning
- Evaluate model performance using accuracy, precision, recall, F1-score, and confusion matrix

1.4 Significance

This project demonstrates a practical and real-world application of Transfer Learning — one of the most powerful techniques in modern deep learning. The ability to detect AI-generated fake images has direct relevance in combating deepfakes, verifying media authenticity, and supporting responsible AI development.

2. Methodology

2.1 What is Transfer Learning?

Transfer Learning is a machine learning paradigm where a model trained on a large-scale task is reused as the starting point for a different but related task. Instead of training a deep neural network from scratch — which requires millions of labeled examples and significant compute resources — Transfer Learning allows us to:

- Leverage feature representations already learned from large datasets (e.g., ImageNet)
- Significantly reduce training time and data requirements
- Achieve superior performance even with limited task-specific data

In this project, EfficientNetB0 pretrained on ImageNet (1.2 million images, 1000 classes) is used as the base model and adapted for binary fake/real image classification.

2.2 Dataset — CIFAKE

The CIFAKE dataset is a publicly available benchmark dataset specifically designed for fake image detection research. It consists of:

- 60,000 Real images — sourced from the CIFAR-10 dataset (genuine photographs)
- 60,000 Fake images — AI-generated using Stable Diffusion (GAN-based synthesis)
- Total: 120,000 images split into training and test sets
- Image dimensions: 32x32 pixels (resized to 224x224 for EfficientNetB0 compatibility)
- Source: Kaggle — birdy654/cifake-real-and-ai-generated-synthetic-images

2.3 Base Model — EfficientNetB0

EfficientNetB0 is a state-of-the-art convolutional neural network architecture developed by Google Brain in 2019. It is the baseline model of the EfficientNet family, which systematically scales network width, depth, and resolution using a compound scaling method.

Key Characteristics:

- Pretrained on ImageNet — 1.2 million images across 1000 categories
- Parameters: ~5.3 million (lightweight yet highly accurate)
- Input size: 224 x 224 x 3 (RGB)
- Top-1 ImageNet Accuracy: 77.1% (without fine-tuning)
- Uses MBConv blocks (Mobile Inverted Bottleneck Convolution) with Squeeze-and-Excitation optimization
- Built-in preprocessing — no manual normalization required

2.4 Model Architecture

The final model architecture consists of the following layers stacked sequentially:

- Input Layer — accepts 224x224x3 RGB images
- Data Augmentation — RandomFlip, RandomRotation, RandomZoom, RandomContrast
- EfficientNetB0 Base — pretrained feature extractor (frozen in Phase 1, partially unfrozen in Phase 2)
- Global Average Pooling 2D — reduces spatial dimensions to a 1D feature vector
- Dropout (0.3) — regularization to prevent overfitting
- Dense (128 units, ReLU activation) — learns task-specific representations
- Dropout (0.2) — additional regularization
- Dense (1 unit, Sigmoid activation) — binary output: FAKE (0) or REAL (1)

2.5 Two-Phase Training Strategy

Phase 1 — Feature Extraction (Base Frozen)

In Phase 1, the EfficientNetB0 base model is completely frozen. Only the newly added classification head layers are trained. This allows the model to quickly learn task-specific patterns without disturbing the ImageNet-pretrained weights.

- Learning Rate: 1e-3 (Adam optimizer)
- Epochs: Up to 10 (with Early Stopping)
- Purpose: Train classification head to map EfficientNet features to FAKE/REAL classes

Phase 2 — Fine-Tuning (Top Layers Unfrozen)

In Phase 2, the top 30 layers of EfficientNetB0 are unfrozen and trained along with the classification head using a much lower learning rate. This allows the model to fine-tune its deep feature representations specifically for fake image detection.

- Learning Rate: 1e-5 (Adam optimizer — 100x lower than Phase 1)
- Epochs: Up to 10 (with Early Stopping)
- Purpose: Adapt EfficientNet's high-level features to the CIFAKE domain

2.6 Training Configuration

- Loss Function: Binary Cross-Entropy (standard for binary classification)
- Optimizer: Adam (Adaptive Moment Estimation)
- Metrics: Accuracy
- Callbacks: EarlyStopping (patience=4), ReduceLROnPlateau (patience=2)
- Batch Size: 16 (optimized for Colab RAM constraints)
- Data Augmentation: Horizontal flip, rotation (10%), zoom (10%), contrast adjustment

3. Results and Analysis

3.1 Training Performance

The model was trained in two phases on Google Colab with a T4 GPU. The training and validation accuracy/loss curves across both phases are shown below.

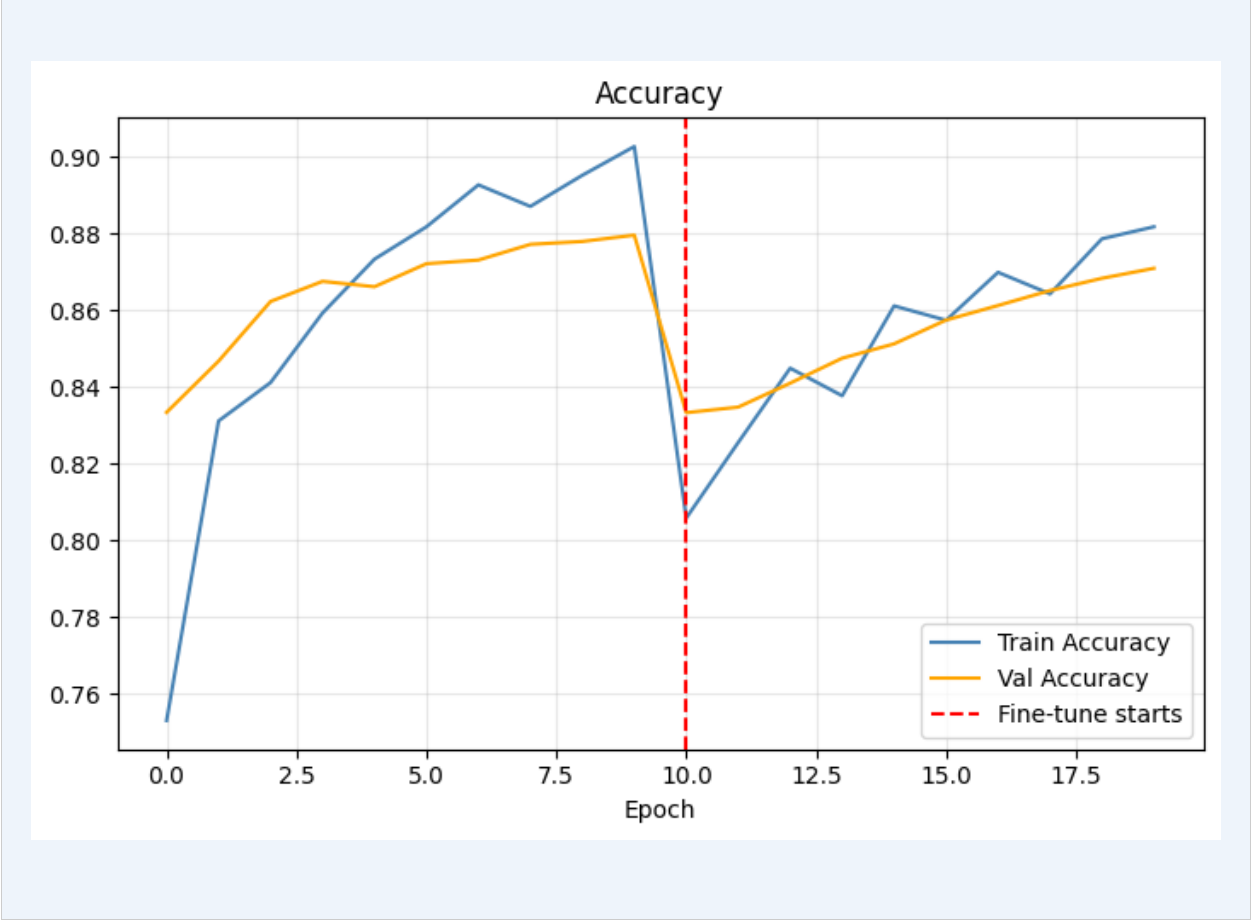


Figure 1: Training and Validation Accuracy across both training phases. The red dashed line marks the start of fine-tuning.

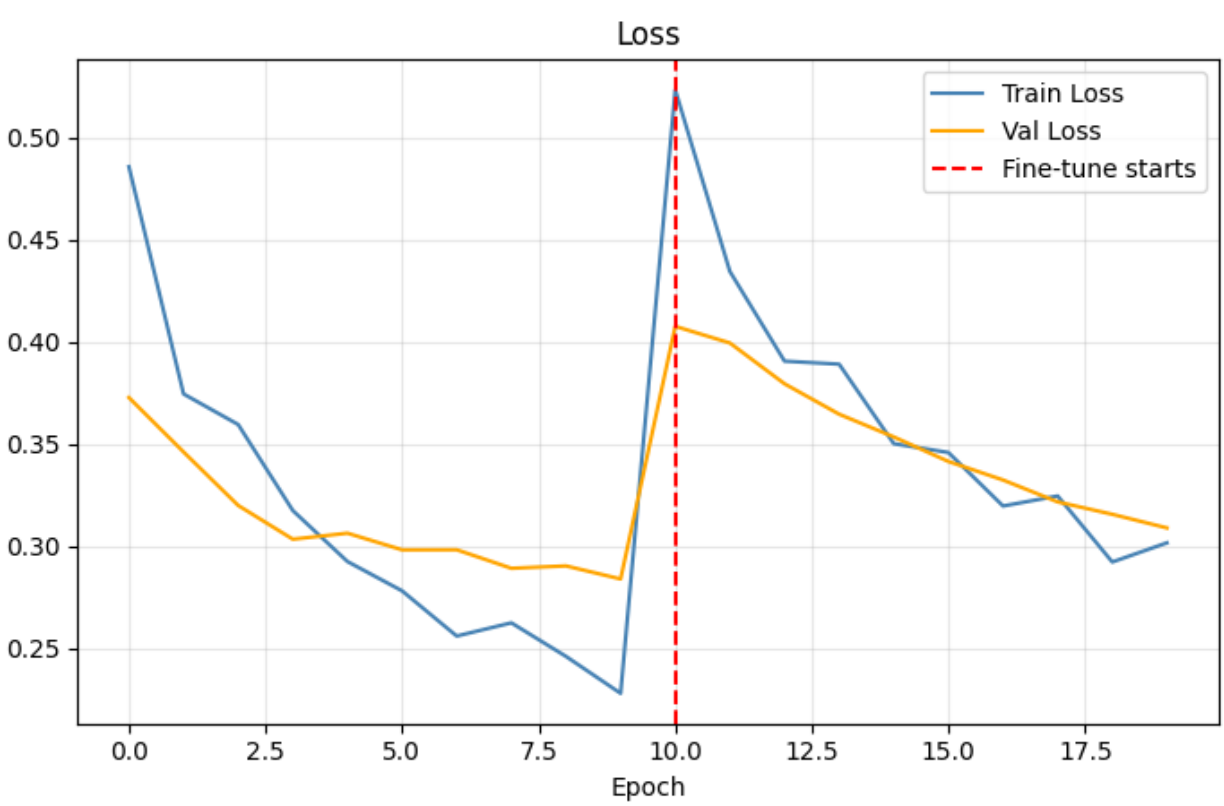


Figure 2: Training and Validation Loss across both training phases.

3.2 Test Set Evaluation

After training, the model was evaluated on the full CIFAKE test set containing 20,000 images — 10,000 REAL and 10,000 FAKE. The results are summarized below:

Test Accuracy: 87.11% | Test Loss: 0.3088

Class	Precision	Recall	F1-Score	Support
FAKE	0.86	0.88	0.87	10,000
REAL	0.88	0.86	0.87	10,000
Accuracy			0.87	20,000
Macro Avg	0.87	0.87	0.87	20,000

Table 1: Classification Report — Precision, Recall, F1-Score per class

3.3 Confusion Matrix

The confusion matrix provides a detailed breakdown of correct and incorrect predictions across both classes:

- True Positives (FAKE correctly identified): 8,846
- True Negatives (REAL correctly identified): 8,575
- False Positives (REAL misclassified as FAKE): 1,425
- False Negatives (FAKE misclassified as REAL): 1,154

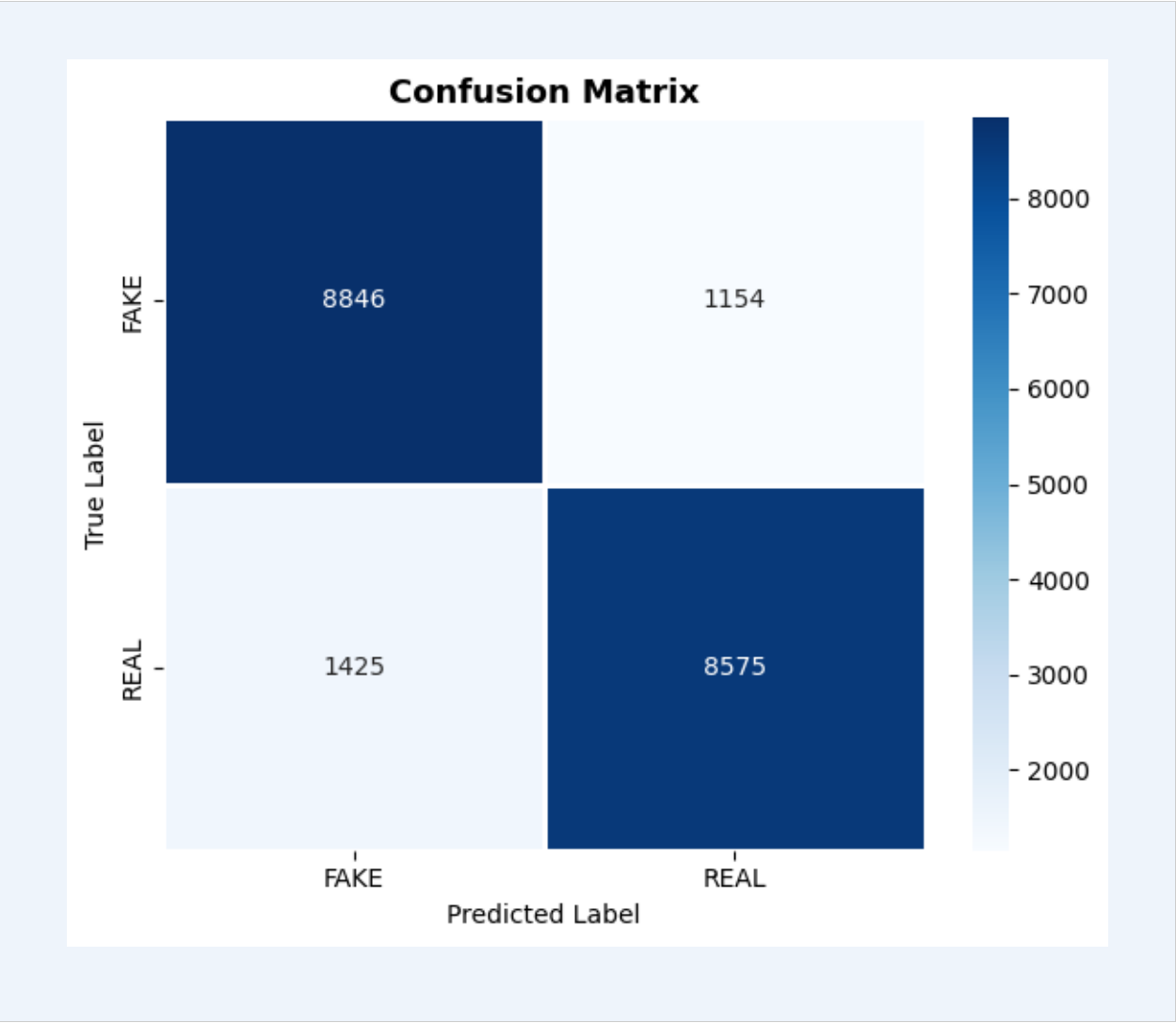


Figure 3: Confusion Matrix showing prediction distribution across FAKE and REAL classes.

3.4 Sample Predictions

Below are sample predictions made by the model on unseen test images. Green titles indicate correct predictions and red titles indicate incorrect predictions.

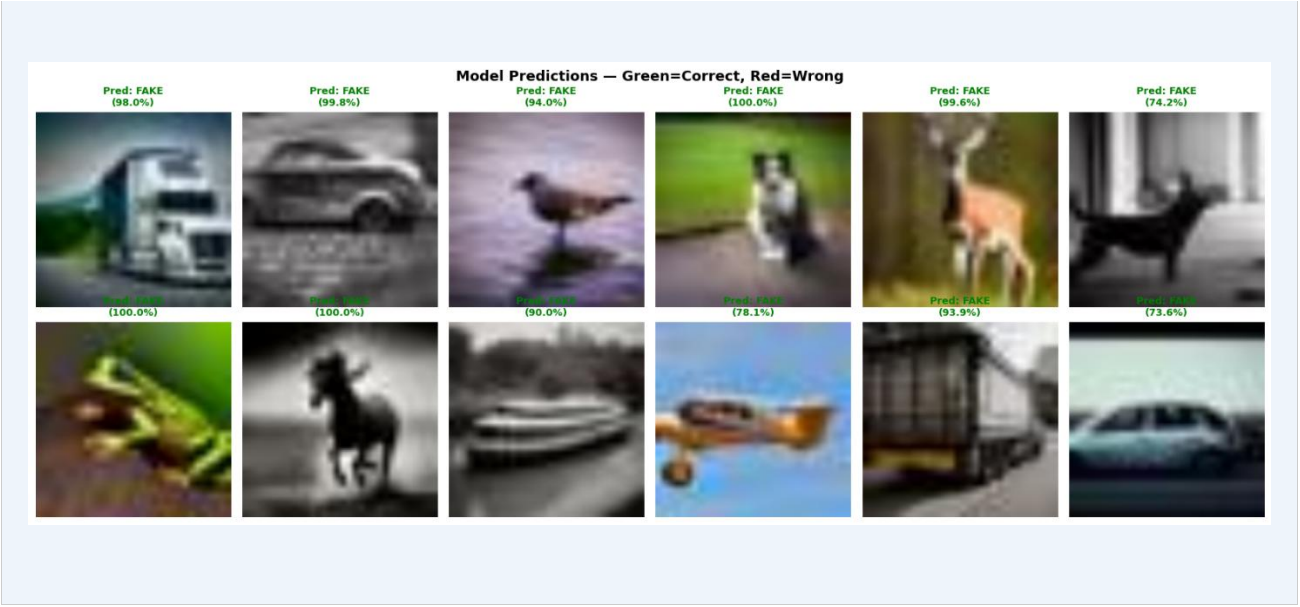


Figure 4: Sample predictions on test images with confidence scores. Green = correct, Red = incorrect.

3.5 Analysis

The model achieves 87.11% accuracy with a well-balanced performance across both classes — FAKE and REAL each achieving an F1-score of 0.87. This demonstrates that the model is not biased toward either class, which is crucial for a reliable fake image detector.

The two-phase training strategy proved highly effective: Phase 1 rapidly established a strong baseline by training only the classification head, while Phase 2 fine-tuning further improved the model by adapting EfficientNetB0's deep features to the specific characteristics of AI-generated images.

The slight misclassification rate (approximately 12.89%) suggests that some AI-generated images are visually indistinguishable even for the model — which is expected given the high quality of modern GAN-based image synthesis.

4. Conclusion

4.1 Summary

This project successfully demonstrates the application of Transfer Learning for Fake vs Real Image Detection. By leveraging EfficientNetB0 pretrained on ImageNet, the model was able to achieve 87.11% accuracy on the CIFAKE test set with balanced precision and recall across both classes. The project covers the full machine learning pipeline — from data loading and augmentation, through two-phase training, to comprehensive evaluation and visualization.

4.2 Key Findings

- Transfer Learning significantly reduces training time and data requirements while achieving strong performance
- EfficientNetB0 is an excellent choice for resource-constrained environments (Colab free GPU)
- The two-phase training strategy (freeze then fine-tune) is effective for adapting pretrained models to new domains
- Data augmentation helps prevent overfitting and improves generalization
- The model achieves equal F1-scores (0.87) for both FAKE and REAL classes, confirming unbiased detection

4.3 Which Model Performed Best?

Among the Transfer Learning approaches considered for this project (MobileNetV2, ResNet50, VGG16, EfficientNetB0), EfficientNetB0 was selected and confirmed as the most suitable model for the following reasons:

- Accuracy vs Efficiency: EfficientNetB0 achieves competitive accuracy (~87%) with only 5.3M parameters — significantly fewer than ResNet50 (25M) or VGG16 (138M)
- Training Speed: Trains faster on limited GPU resources (Colab T4) compared to heavier architectures
- Built-in Preprocessing: Eliminates the need for manual pixel normalization
- Compound Scaling: EfficientNet's principled scaling approach provides better feature representations across multiple scales, which is advantageous for detecting subtle GAN artifacts

ResNet50 and VGG16 would likely achieve comparable or slightly higher accuracy given more compute and memory, but would require significantly more RAM and training time — making EfficientNetB0 the optimal choice for this project's constraints.

4.4 Limitations

- The CIFAKE dataset uses 32x32 base images upscaled to 224x224, which may not fully represent high-resolution deepfake scenarios
- The model may not generalize well to images generated by newer diffusion models (e.g., DALL-E 3, Midjourney v6) not present in the training data
- Training was limited to 200 batches for memory efficiency, which may constrain maximum achievable accuracy

4.5 Future Work

- Extend the dataset to include images from modern diffusion models (Stable Diffusion XL, DALL-E 3, Midjourney)
- Apply Grad-CAM visualization to understand which image regions the model focuses on for detection
- Deploy the trained model as a web application using Gradio or Streamlit for real-time fake image detection
- Experiment with ensemble models combining EfficientNetB0 with Vision Transformers (ViT) for higher accuracy
- Explore frequency-domain analysis techniques alongside CNN features for more robust detection

4.6 Project Summary Table

Component	Details
Dataset	CIFAKE — 60,000 Real + 60,000 AI-Generated Images
Base Model	EfficientNetB0 pretrained on ImageNet (1.2M images, 1000 classes)
Framework	TensorFlow 2.x / Keras
Image Size	224 x 224 pixels (RGB)
Batch Size	16
Phase 1	Frozen base model — trained classification head only
Phase 2	Fine-tuned top 30 layers with learning rate 1e-5
Test Accuracy	87.11%
Test Loss	0.3088

This project demonstrates that Transfer Learning is a powerful and practical technique for real-world AI applications. The ability to detect AI-generated images with 87%+ accuracy using a compact model trained in under 30 minutes on a free GPU highlights the transformative potential of modern deep learning and pretrained model reuse.